



Modelling Financial Markets

Seminar 17: Bubbles and Crashes — Hands-on



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Acronyms used in this material

- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-FI: BET Financial Investment companies index
- ▣ BIC: Bayesian Information Criterion
- ▣ BSADF: Backward Sup Augmented Dickey–Fuller (statistic)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ GSADF: Generalised Sup Augmented Dickey–Fuller (test)
- ▣ LPPLS: Log-Periodic Power Law Singularity (model)
- ▣ LR: Likelihood Ratio (test)
- ▣ NVIDIA: NVIDIA Corporation (company name, US chip designer)
- ▣ PSY: Phillips–Shi–Yu (bubble test and date-stamping procedure)
- ▣ PWY: Phillips–Wu–Yu (bubble test and date-stamping procedure)
- ▣ SADF: Sup Augmented Dickey–Fuller (test)
- ▣ SE: Standard Error

Outline

Introduction

Part A: Pen and Paper

Part B: Estimation and Inference

Part C: Open Question

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Separate fundamental value and bubble in a present-value model
- ▣ Compute a right-tailed ADF statistic step by step and run GSADF with simulated and bootstrap critical values
- ▣ Fit an LPPLS model and judge how stable its critical time is
- ▣ Estimate a Markov-switching model and read filtered and smoothed probabilities

- ▣ Background: Lecture 17; textbook [Franke, Härdle & Hafner \(2019\)](#)

Structure

- ▣ **Part A:** eight pen-and-paper exercises (A1–A8)
- ▣ **Part B:** eight computational exercises with inference (B1–B8)
- ▣ **Part C:** one open question, a seed for the team project (C1), and the critique of an AI answer (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- Explosiveness tests: log levels, no lagged terms, minimum window $r_0 = 0.01 + 1.8/\sqrt{T}$, minimum duration $\log T$
- Always ask: would this signal have been available at the time, and in which direction did prices move?

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (fundamental value and bubble), A3 (right-tailed ADF step by step), A5 (Markov switching: durations and one filter step), A7 (LPPLS diagnostics)
- Part B: B1 (GSADF on the price-dividend ratio), B3 (Bitcoin: Monte Carlo vs wild bootstrap), B5 (dispersion of the LPPLS critical time), B7 (Markov switching on the S&P 500)

Proposed: you solve them

- Part A: A2 (Blanchard–Watson; model: A1), A4 (PSY windows; model: A3), A6 (filter step; model: A5), A8 (LPPLS filter; model: A7)
- Part B: B2 (SADF vs GSADF; model: B1), B4 (BET and BET-FI; model: B1, B3), B6 (confidence indicator; model: B5), B8 (Bitcoin regimes; model: B7)
- Part C: C1, is there an AI bubble?; C2, critique an AI answer on SADF and LPPLS

Setup: Python Environment and Data

-  **Open in Colab:** [chapter17_seminar_notebook.ipynb](#)
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `statsmodels`, `matplotlib`, `xlrd`
- Series: S&P Composite real price and dividend (monthly, 1871–2024, Robert J. Shiller); Nasdaq 100, S&P 500, BET, BET-FI; Bitcoin; NVIDIA, Cisco; closes up to 18 September 2026
- Weekly and monthly series: last close of the week or month; stocks adjusted for dividends and splits

Formulas Used Today

- Fundamental value (Gordon): $F = D_1/(r - g)$; rational bubble: $\mathbb{E}_t[B_{t+1}] = (1 + r)B_t$ Blanchard & Watson (1982)
- Right-tailed ADF: $\Delta y_t = a + b y_{t-1} + \varepsilon_t$, $\hat{b} = S_{xz}/S_{xx}$ (centred sums), $\text{SE}(\hat{b}) = \sqrt{s^2/S_{xx}}$, $s^2 = \text{SSR}/(n - 2)$
- GSADF = $\sup_{r_2} \sup_{r_1} \text{ADF}'_{r_1}{}^2$; BSADF $_{r_2} = \sup_{r_1} \text{ADF}'_{r_1}{}^2$ Phillips, Shi & Yu (2015a)
- Markov switching: duration $1/(1 - p_{ii})$; stationary $\pi_2 = p_{12}/(p_{12} + p_{21})$; filter $P(s_t \mid \text{data}_t) \propto P(s_t \mid \text{data}_{t-1}) \varphi(r_t; \mu_{s_t}, \sigma_{s_t})$ Hamilton (1989)
- LPPLS: $\ln p = A + B(t_c - t)^m + (t_c - t)^m [C_1 \cos(\omega \ln(t_c - t)) + C_2 \sin(\omega \ln(t_c - t))]$; $O = \frac{\omega}{2\pi} \ln \frac{t_c - t_1}{t_c - t_2}$, $D = \frac{m|B|}{\omega|C|}$ Filimonov & Sornette (2013)

A1: Fundamental Value and a Rational Bubble [Solved]

Task

- A share pays next year a dividend $D_1 = 2$ that grows at $g = 2\%$; required return $r = 6\%$; market price $P = 65$
- (a) Fundamental value (Gordon) and bubble $B = P - F$
- (b) Expected F and B in 5 and 30 years, if the bubble is rational
- (c) What happens to the bubble share of the price?

Solution

- $F = 2/(0.06 - 0.02) = 50.0$, $B = 65 - 50.0 = 15.0$, share 23.1%
- Fundamental grows at g , bubble at r :
 $F_5 = 50.0 \cdot 1.02^5 = 55.2$, $\mathbb{E}[B_5] = 15.0 \cdot 1.06^5 = 20.1$
- $F_{30} = 90.6$, $\mathbb{E}[B_{30}] = 86.2$
- Bubble share: 23.1% today, 26.7% in 5 years, 48.8% in 30 years
- Since $r > g$ the bubble eventually dominates the price: its path is explosive relative to fundamentals

A2: A Bubble That Can Burst [Proposed]

Solution discussed in class

Task

- The bubble of A1 ($B_0 = 15$, $r = 6\%$) survives each year with probability $\pi = 0.9$ and bursts otherwise [Blanchard & Watson \(1982\)](#)
- (a) Growth rate while it survives
- (b) Probability that it survives 5 years; its expected lifetime
- (c) Its value in 5 years if it survives, and its expected value
- Model: A1

A3: Right-Tailed ADF, Step by Step [Solved]

Task

- ▣ Monthly log prices: 4.60, 4.62, 4.66, 4.71, 4.79, 4.90, 5.05, 5.25
- ▣ (a) Regression $\Delta y_t = a + b y_{t-1} + \varepsilon_t$ on the $n = 7$ changes: $\hat{b}$ and $\text{SE}(\hat{b})$
- ▣ (b) The ADF statistic and the decision at 5% against $H_1 : b > 0$
- ▣ (c) Why is the critical value not 1.645?

Solution

- ▣ $\bar{x} = 4.7614$, $\bar{z} = 0.0929$; $S_{xx} = \sum (x - \bar{x})^2 = 0.1623$,
 $S_{xz} = \sum (x - \bar{x})(z - \bar{z}) = 0.06357$
- ▣ $\hat{b} = S_{xz}/S_{xx} = 0.3917$, $\hat{a} = \bar{z} - \hat{b}\bar{x} = -1.7723$
- ▣ $\text{SSR} = 2.40 \cdot 10^{-4}$, $s^2 = \text{SSR}/5 = 4.806 \cdot 10^{-5}$,
 $\text{SE}(\hat{b}) = \sqrt{s^2/S_{xx}} = 0.0172$
- ▣ $\text{ADF} = 0.3917/0.0172 = 22.76$ vs the simulated 95% critical value for 7 observations 0.24 (99%: 1.12): reject the random walk
- ▣ Under H_0 (unit root) the t -statistic is not Student- t : its distribution is Dickey-Fuller, shifted to the left

A4: How Many Windows? [Proposed]

Solution discussed in class

Task

- Monthly Nasdaq 100, $T = 440$ observations Phillips, Shi & Yu (2015a)
- (a) Minimum window fraction $r_0 = 0.01 + 1.8/\sqrt{T}$ and length $w_0 = \lfloor r_0 T \rfloor$
- (b) Number of ADF regressions in SADF and in GSADF
- (c) Minimum duration $\log T$; repeat (a) and (c) for 10 years of daily data ($T = 2520$)
- Model: A3

A5: Markov Switching: Durations and One Filter Step [Solved]

Task

- Weekly returns (%): calm $\mu_1 = 0.3$, $\sigma_1 = 2$; turbulent $\mu_2 = -0.5$, $\sigma_2 = 5$; $p_{11} = 0.98$, $p_{22} = 0.90$ Hamilton (1989)
- (a) Expected durations and stationary probabilities
- (b) Unconditional mean and standard deviation of the return
- (c) Last week $P(\text{turbulent}) = 0.2$; this week $r_t = -6\%$: update the probability

Solution

- Durations $1/0.02 = 50$, $1/0.10 = 10$ weeks;
 $\pi_2 = 0.02/(0.02 + 0.10) = 0.167$
- $\mathbb{E}[r] = 0.833 \cdot 0.3 + 0.167 \cdot (-0.5) = 0.167$;
 $\text{Var}(r) = \sum \pi_i \sigma_i^2 + \sum \pi_i (\mu_i - \mathbb{E}r)^2 = 7.59$, s.d. 2.75
- Prediction: $0.2 \cdot 0.90 + 0.8 \cdot 0.02 = 0.196$
- Densities $\varphi(-6; 0.3, 2) = 0.00140$,
 $\varphi(-6; -0.5, 5) = 0.04357$
- Posterior: $\frac{0.196 \cdot 0.04357}{0.196 \cdot 0.04357 + (1 - 0.196) \cdot 0.00140} = 0.884$: one bad week almost settles it

A6: Another Filter Step [Proposed]

Solution discussed in class

Task

- ▣ Calm $\mu_1 = 0.4$, $\sigma_1 = 2$; turbulent $\mu_2 = -1$, $\sigma_2 = 6$; $p_{11} = 0.95$, $p_{22} = 0.80$
- ▣ (a) Durations, stationary probabilities, unconditional mean and s.d.
- ▣ (b) Last week $P(\text{turbulent}) = 0.1$; this week $r_t = +1\%$: update
- ▣ (c) Probability of the turbulent regime two weeks ahead
- ▣ Model: A5

A7: Reading an LPPLS Fit [Solved]

Task

- A fit on $[t_1, t_2] = [2025.0, 2026.0]$ (years) gives $t_c = 2026.08$, $m = 0.5$, $\omega = 8$, $B = -1.2$, $C_1 = 0.06$, $C_2 = -0.02$ Filimonov & Sornette (2013)
- (a) Amplitude C , number of oscillations O , damping D
- (b) Does it pass the filter ($B < 0$; $m \in [0.01, 0.99]$; $\omega \in [2, 15]$; $t_c - t_2 \leq 0.1(t_2 - t_1)$; $O \geq 2.5$; $D \geq 0.5$)?
- (c) Trend growth rate of the log price at t_2

Solution

- $C = \sqrt{0.06^2 + 0.02^2} = 0.0632$
- $O = \frac{8}{2\pi} \ln \frac{1.08}{0.08} = \frac{8}{2\pi} \cdot 2.603 = 3.31$; $D = \frac{0.5 \cdot 1.2}{8 \cdot 0.0632} = 1.19$
- All six conditions hold ($t_c - t_2 = 0.08 \leq 0.1$, i.e. 29 days): a qualified fit
- Trend growth
 $-Bm(t_c - t_2)^{m-1} = 1.2 \cdot 0.5 \cdot 0.08^{-0.5} = 2.12$ log points per year, and rising

A8: A Fit That Fails the Filter [Proposed]

Solution discussed in class

Task

- ▣ Same window; now $t_c = 2026.45$, $m = 0.95$, $\omega = 3.5$, $B = -0.8$, $C_1 = 0.25$, $C_2 = 0$
- ▣ (a) Compute O and D
- ▣ (b) Which filter conditions fail, and what does each failure mean?
- ▣ Model: A7

B1: GSADF on the S&P 500 Price-Dividend Ratio [Solved]

Task

- Log real price-dividend ratio, monthly 1871–2024 ($T = 1841$)
- (a) GSADF with the PSY minimum window and 1 000 Monte Carlo paths [Phillips, Shi & Yu \(2015a\)](#)
- (b) Repeat with $r_0 = 0.03$ and $r_0 = 0.08$; date-stamp the episodes each time
- Code: `psy`, `psy_cv`, `episodes` in the notebook

Interpretation

- Which episodes survive every window choice?
- Why can a very short minimum window produce spurious episodes?
- Is every explosive episode of the ratio a bubble?

B1: Results [Solved]

r_0	w_0	GSADF	cv 95%	Episodes (BSADF)
0.030	55	3.93	2.55	Jul 1878–Feb 1879; Apr 1879–Apr 1880; Aug 1917–May 1918 (falling); Nov 1928–Sep 1929; Dec 1954–Apr 1956; May 1997–Feb 2001; Sep 2008–Apr 2009 (falling)
0.052	95	3.41	2.39	Aug 1917–May 1918 (falling); May 1997–Jul 2001
0.080	147	1.77	2.24	May 1997–Aug 2001

- The dot-com episode (1997–2001) is date-stamped with every window; with $r_0 = 0.08$ GSADF is below its critical value, yet BSADF still crosses its pointwise critical values: the global test is stricter than the date-stamping
- The shortest window adds 1878–1880, 1928–1929, 1955–1956 and 2008–2009: fewer observations per regression, noisier ADF statistics, higher critical values, more short episodes
- 1917–1918 is a fall of the ratio (real prices dropped faster than real dividends): explosive, but not a bubble

B2: SADF vs GSADF [Proposed]

Solution discussed in class

Task

- Monthly log closes 1990–2026 of the Nasdaq 100 and of the S&P 500
- (a) SADF and GSADF with their 95% critical values
- (b) Date-stamp with the recursive ADF (PWY) and with BSADF (PSY); list the episodes [Phillips, Wu & Yu \(2011\)](#)
- Model: B1

Interpretation

- Which episodes does only one of the two methods find, and why?
- Is any episode an accelerating fall rather than a rise?

B3: Bitcoin: Simulated vs Bootstrap Critical Values [Solved]

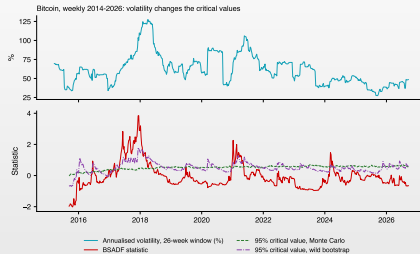
Task

- Weekly log closes of Bitcoin, 2014–2026
- (a) GSADF with Monte Carlo critical values (Gaussian random walk)
- (b) Wild bootstrap: multiply the demeaned weekly changes by random signs, 1 000 times [Harvey et al. \(2016\)](#), [Phillips & Shi \(2020\)](#)
- (c) Date-stamp with both critical-value sequences

Interpretation

- When are the bootstrap critical values higher, and why?
- Which episode is robust to the choice of critical values?

B3: Results [Solved]



- GSADF = 3.85; 95% critical value: Monte Carlo 2.27, wild bootstrap 2.82: explosive in both cases
- Monte Carlo episodes: Apr 2017–Jun 2018; Dec 2020–May 2021
 - ▶ Wild bootstrap episodes: May 2017–Jan 2018; Dec 2020–Feb 2021
- The bootstrap critical values rise in and after the volatile periods (up to 1.80 vs 0.71): both episodes survive, but shorter

B4: The Bucharest Stock Exchange [Proposed]

Solution discussed in class

Task

- BET, monthly 1997–2026; BET-FI (financial investment companies), weekly 2012–2026
- (a) GSADF with Monte Carlo and wild bootstrap critical values
- (b) Date-stamp the episodes; classify each as a rise or a fall
- Model: B1, B3

Interpretation

- Was the 2003–2007 boom explosive in real time? When did the signal start? [Pele, Mazurencu & Nijkamp \(2013\)](#)
- Is the current rise of both indices explosive, and does that make it a bubble?

B5: How Much Does the Critical Time Depend on the Window? [Solved]

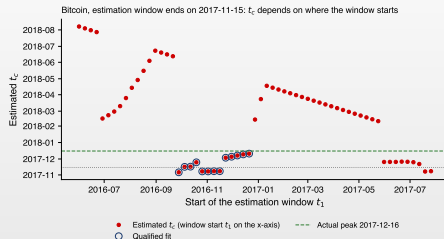
Task

- Bitcoin daily log prices; windows ending on 15 November 2017 and starting every 7 days between June 2016 and August 2017
- (a) Fit the LPPLS model on each window (Filimonov–Sornette) [Filimonov & Sornette \(2013\)](#)
- (b) Distribution of t_c ; share of qualified fits; share of fits at the edge of the search ranges

Interpretation

- Would you trust a single t_c reported without the window?
- Do the qualified fits point closer to the actual peak?

B5: Results [Solved]



- ▣ 61 windows: t_c between 5 Nov 2017 and 7 Aug 2018, median 27 Feb 2018; interquartile range of the error 133 days; actual peak 16 Dec 2017
- ▣ Qualified fits: 21%; fits at the edge of the ranges of m or ω : 26%; within 30 days of the peak: 25%
- ▣ The qualified fits cluster between early November and mid-December 2017, just before the peak; the others spread over the following 9 months

B6: The Confidence Indicator on the Nasdaq 100 [Proposed]

Solution discussed in class

Task

- Nasdaq 100, daily; indicator every 2 weeks from 1997: share of 11 windows (90–690 days) with a qualified fit [Demirer et al. \(2019\)](#)
- (a) When was the indicator first positive, and how often in 1999–March 2000?
- (b) Compare the frequency of a 20% fall within 90 days after a signal and without a signal
- Model: B5

Interpretation

- Would the indicator have helped an investor in 2000?
- How would you test whether the difference in (b) is significant?

B7: Markov Switching on the S&P 500 [Solved]

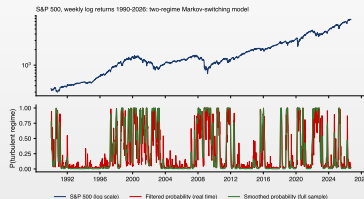
Task

- Weekly log returns of the S&P 500, 1990–2026 (1915 weeks); two regimes with switching mean and variance [Hamilton \(1989\)](#)
- (a) Estimates with standard errors; expected durations
- (b) First week of 2000, of autumn 2008 and of 2020 with a filtered turbulent probability above 0.5
- (c) How often do filtered and smoothed regimes agree?

Interpretation

- Is the turbulent mean significantly negative?
- The likelihood-ratio test of one vs two regimes is non-standard: why?

B7: Results [Solved]



Regime	$\hat{\mu}$	SE	$\hat{\sigma}$	$\hat{p}_{ij}$	Duration
Calm	0.32	0.04	1.49	0.974	38.1
Turbulent	-0.27	0.17	3.67	0.931	14.4

- First week above 0.5: 28 Jan 2000 (2000), 3 Oct 2008 (autumn 2008), 28 Feb 2020 (2020); filtered and smoothed regimes agree in 90% of weeks
- Turbulent mean: $t = -0.27/0.17$, not significant; $LR = 2(-4070.0 - (-4326.0)) = 512.1$, but the regime parameters are not identified under one regime, so χ^2 critical values do not apply

B8: Bitcoin: Two or Three Regimes? [Proposed]

Solution discussed in class

Task

- Weekly log returns of Bitcoin, 2014–2026 (626 weeks)
- (a) Fit two and three regimes with switching mean and variance
- (b) Compare them with AIC and BIC; describe each regime (mean, volatility, duration)
- Model: B7

Interpretation

- Which regime do the 2017 and 2021 booms belong to?
- Is the extra regime worth its parameters?

C1: Is There an AI Bubble?

- Compare NVIDIA and the Nasdaq 100 today with Cisco and the Nasdaq 100 in the dot-com episode
- Explosiveness: GSADF/BSADF on monthly and weekly data, Monte Carlo and wild bootstrap critical values
 - ▶ When did each signal start; how far had prices risen by then; is the current episode still open?
- Run-ups à la Greenwood, Shleifer and You: two-year return, raw and net of the Nasdaq 100 [Greenwood, Shleifer & You \(2019\)](#)
- LPPLS: is there a robust super-exponential signature today (confidence indicator)?
- Fundamentals: which data on earnings or sales would you add to separate a bubble from explosive fundamentals [Pástor & Veronesi \(2009\)](#)?
- Deliverable: a two-page note with charts, and a Quantlet folder (code, Metainfo, charts)

C2: Critique an AI Answer (1/2) [Proposed]

Solution discussed in class

- **Student prompt:** “y is the monthly log price-dividend ratio of the S&P 500. Write Python code for SADF with the PSY minimum window, decide at 5% whether there is a bubble, date-stamp it and tell me when it will burst.”
- **Answer of the AI assistant**

```
import numpy as np
from statsmodels.tsa.stattools import adfuller
T = len(y)
w0 = int(np.floor((0.01 + 1.8 / np.sqrt(T)) * T))
adf = lambda x: adfuller(x, maxlag=0, regression='c', autolag=None)[0]
sadf = max(adf(y[:t]) for t in range(w0, T + 1))
cv = adfuller(y, maxlag=0, regression='c', autolag=None)[4]['5%']
print('bubble' if sadf > cv else 'no bubble')
flag = [adf(y[t - w0:t + w0]) > cv for t in range(w0, T - w0)]
```

- “An LPPLS fit on the last 36 months gives $\hat{t}_c = \text{March 2027}$: this is when the bubble bursts.”

C2: Critique an AI Answer (2/2) [Proposed]

Solution discussed in class

- The answer on the previous slide contains **three** errors typical for bubble tests; the rest is correct
 - ▶ SADF: supremum of ADF statistics over forward-expanding windows (Lecture 17)
 - ▶ PSY: Phillips, Shi and Yu, the authors of the GSADF test [Phillips, Shi & Yu \(2015a\)](#)
- Your tasks
 - ▶ Find each error: which line, and why it is wrong
 - ▶ Say how you would detect it without knowing the answer: a test on simulated data, a truncated sample, a second window
 - ▶ Give the corrected code or statement
 - ▶ Log each error as in [AI_ERRORS.md](#): what the AI answered, what was wrong, how you found out, the correct version

Common Pitfalls

- ▣ Using the left-tailed Dickey–Fuller table for a right-tailed test
- ▣ Calling an explosive episode a bubble without checking its direction (up or down)
- ▣ Reading date-stamps from smoothed or full-sample quantities as if they were available in real time
- ▣ Reporting one LPPLS critical time without the window, the filter and its dispersion
- ▣ Ignoring volatility shifts in crypto data: use the wild bootstrap
- ▣ Using weekly or monthly data for short squeezes that last days

Take-Home Assignment (Team Project)

- Choose three assets not covered in class (a BVB stock, a crypto-asset, a sector or country index)
- Run GSADF with Monte Carlo and wild bootstrap critical values at two frequencies; date-stamp and classify the episodes
- Fit LPPLS on the largest episode; report the dispersion of t_c across windows and the confidence indicator
- Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations
- If you use an AI assistant, log every error you catch in [AI_ERRORS.md](#) (as in C2)

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Cryptocurrency: Speculative Asset and Medium of Exchange
- ▣  Statistics of Financial Markets

Code and data

- ▣ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_17

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